

A resolved view of star formation, gas cycling and chemical enrichment in Virgo Cluster spirals.

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Feb 26, 2026



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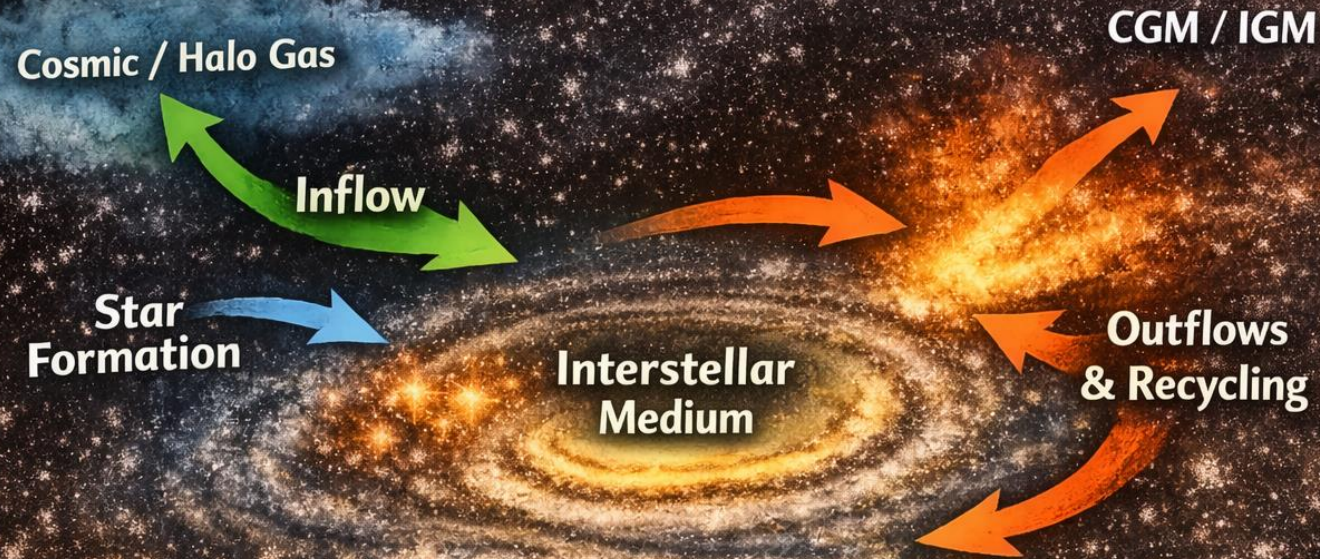
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Galaxy evolution is regulated by the baryon cycle

The Baryon Cycle

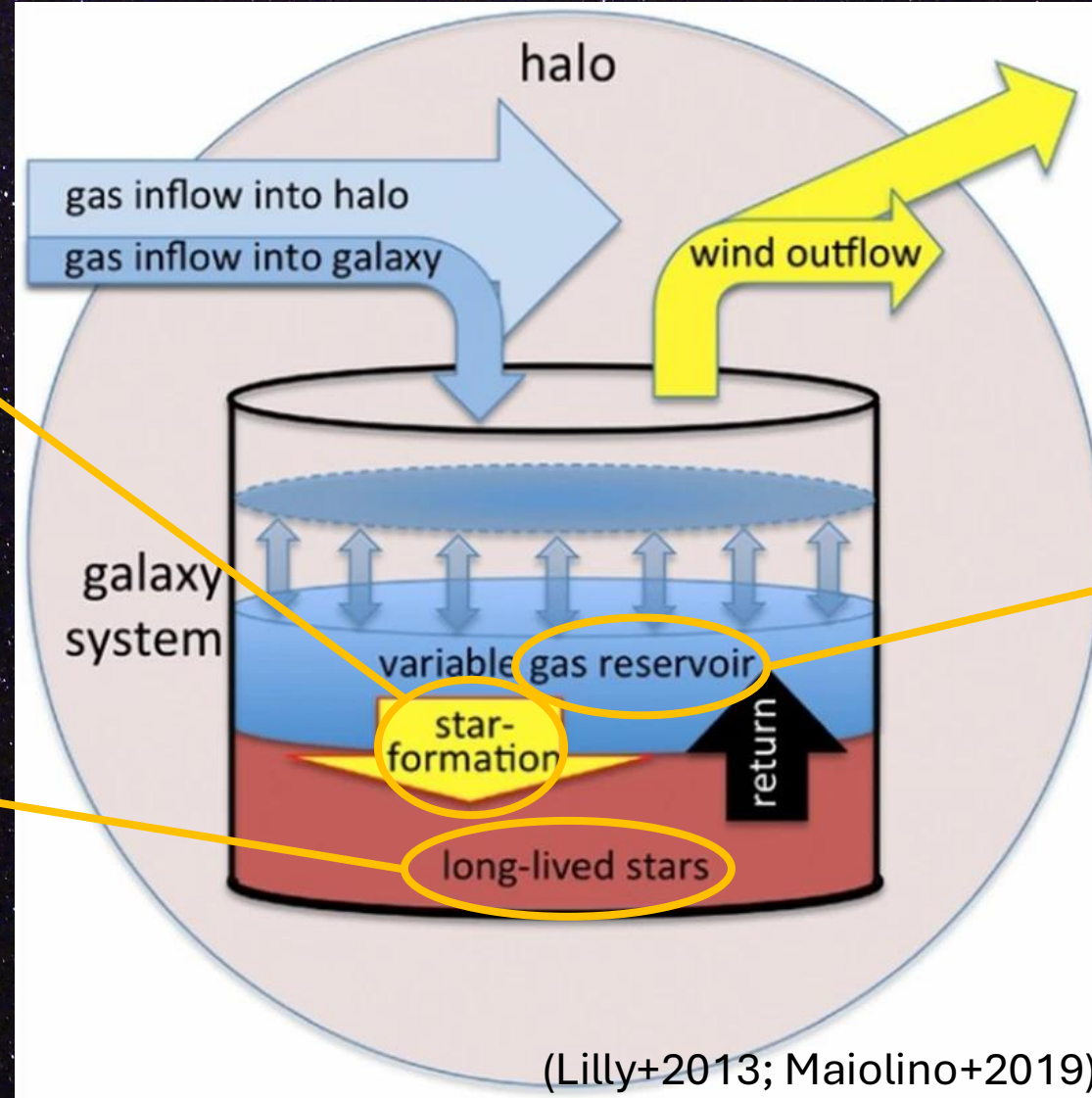


- Gas inflow → fuels star formation
- Star formation → builds stars + metal synthesis
- Feedback → drives outflows + recycling

Bathtub/Gas-regulator Model

Star formation Rate
(SFR) = current growth
rate (“how fast
building now”)

Stellar Mass (M_*) =
integrated past star
formation (“how much
built up”)

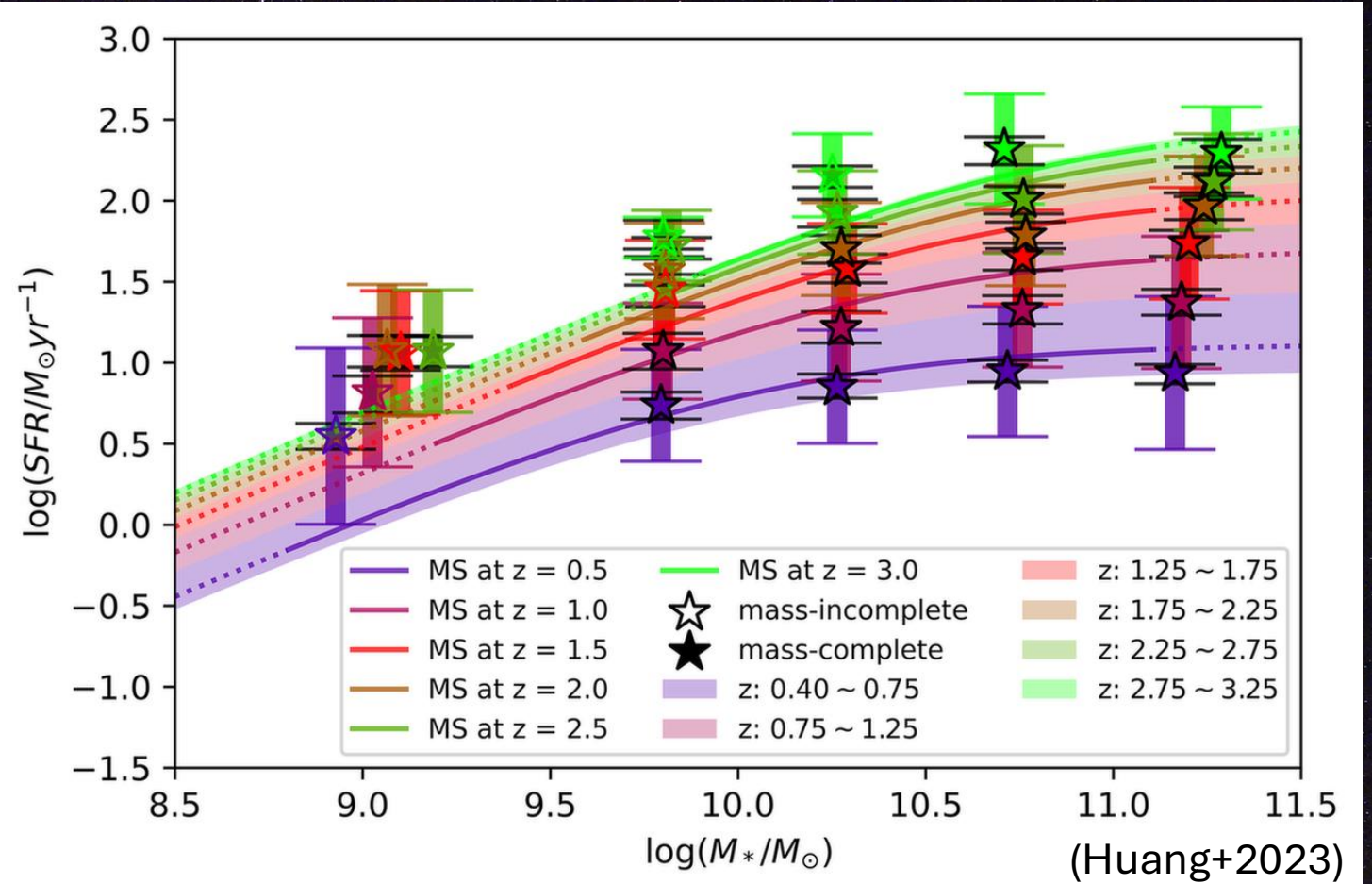


Gas-phase metallicity
(Z_{gas}) = enrichment vs
dilution (“how
processed the gas is”)

Global SFMS (SFR vs M_*)

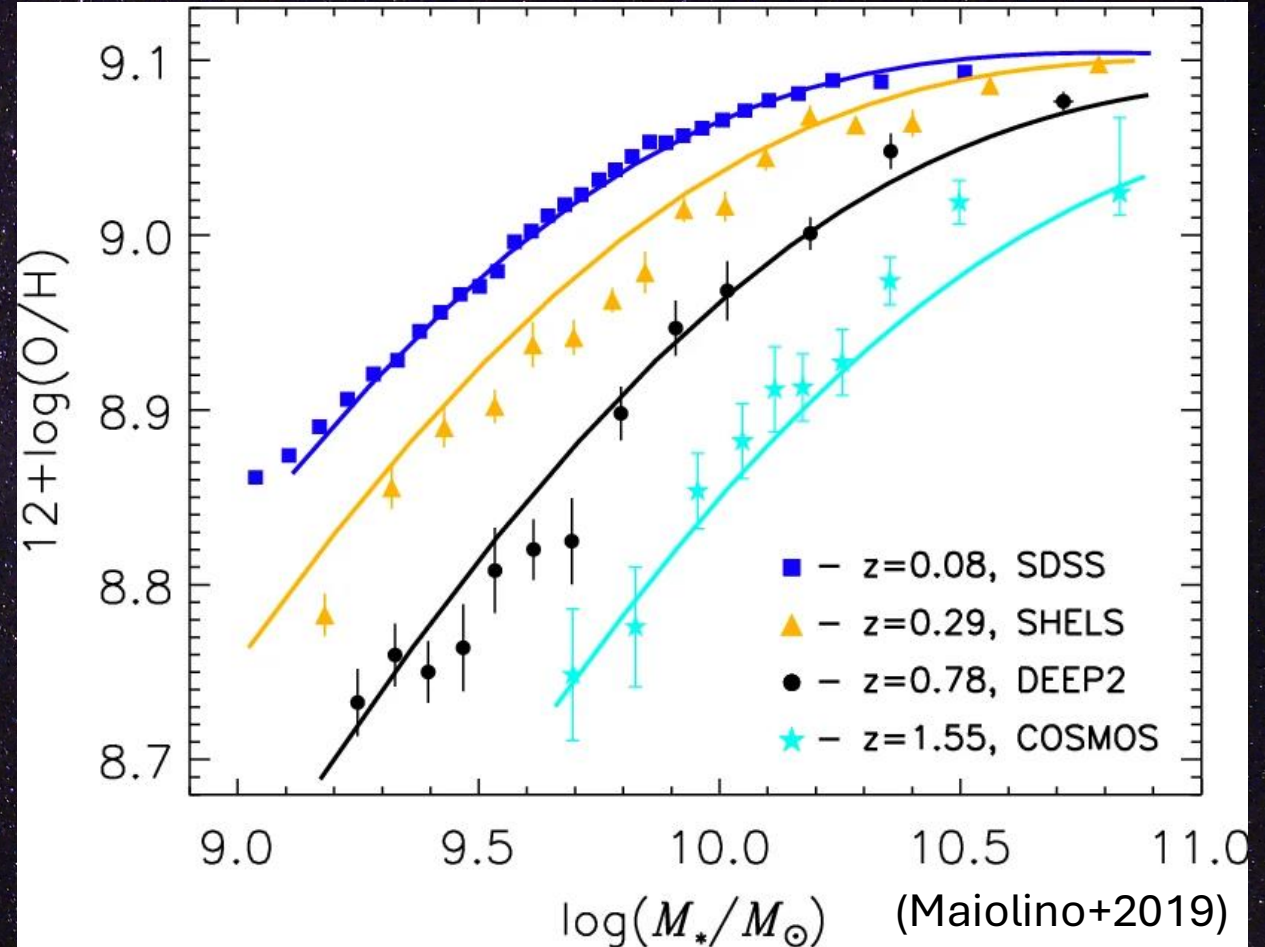
Gas-star formation cycle

- SFR tracks gas accretion and stellar mass build up
- $\Rightarrow$ **Star-Forming Main Sequence (SFMS):** $\text{SFR} \propto M_*$ with small scatter and evolve with redshift



Global MZR (Z_{gas} vs M_*)

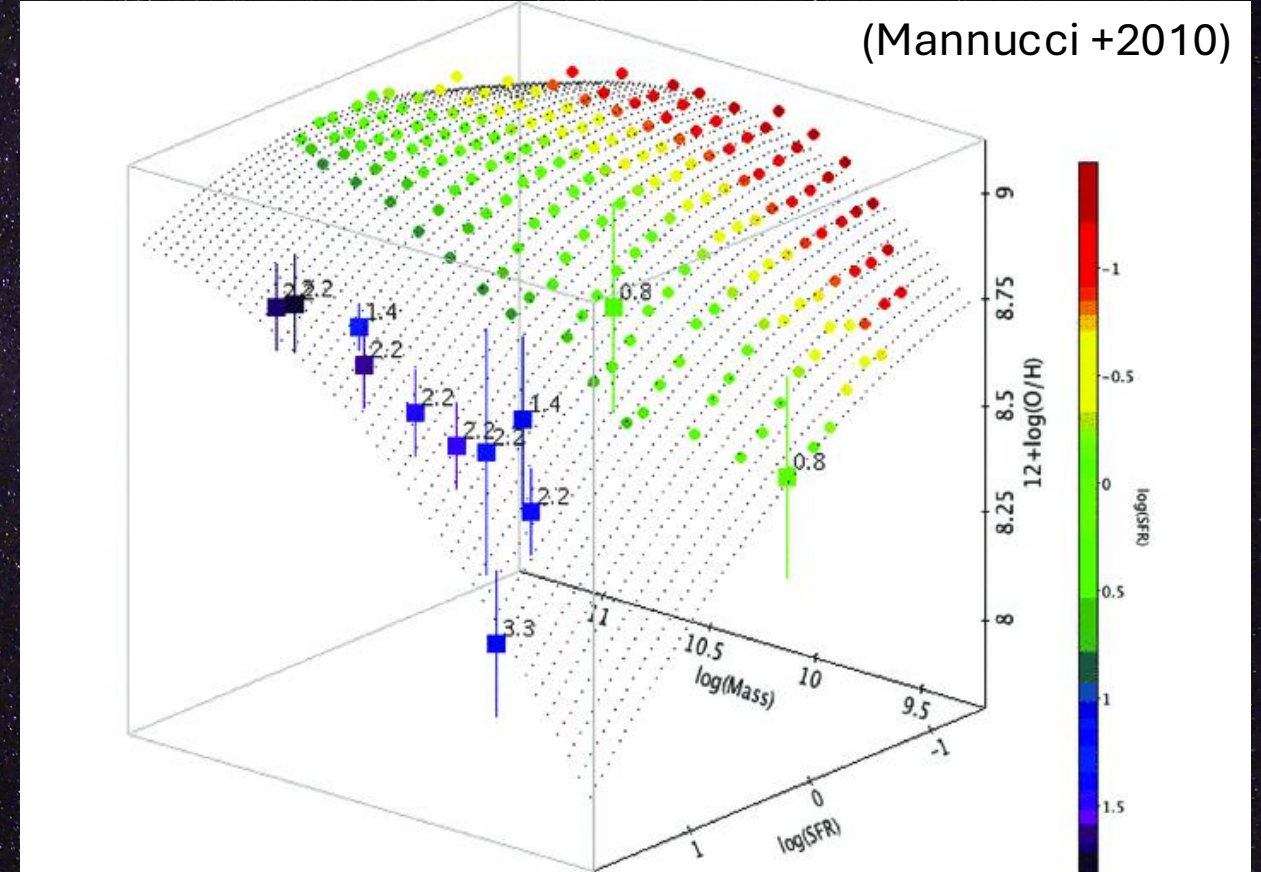
- Metal enrichment by star formation, dilution by inflow, and removal via outflows.
- Deeper potentials (higher M_*) $\Rightarrow$ better metal retention $\Rightarrow$ **Mass-Metallicity Relation (MZR)**: Z_{gas} increases with M_* (steep at low mass, saturating at high mass, and evolve with cosmic time).
- Z_{gas} : Commonly referred to the oxygen abundance, $12+\log(\text{O}/\text{H})$



Global FMR (interplay between M_* , SFR and Z_{gas})

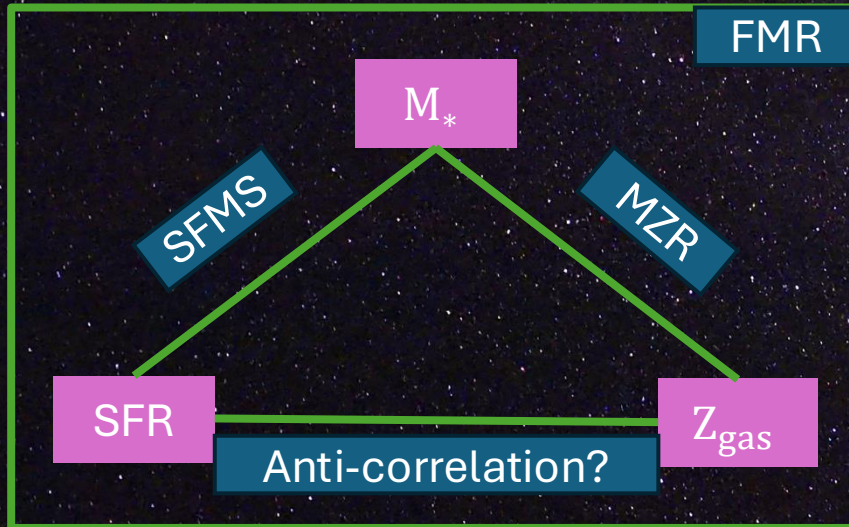
Fundamental Metallicity Relation (FMR)

- A 3D surface: $Z_{\text{gas}}(M_*, \text{SFR})$
- “Fundamental”: smaller scatters than SFMS and MZR; roughly invariant with redshift to $z \sim 2-3$
- Interpretation: Adding SFR as a second dependence of MZR.

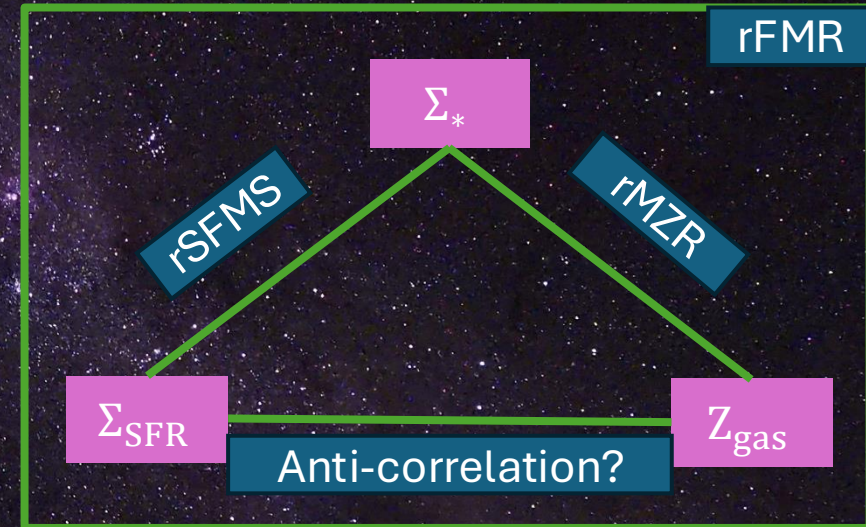


Global (Integrated) $\Rightarrow$ Local (Resolved)

Integrated galaxy-wide

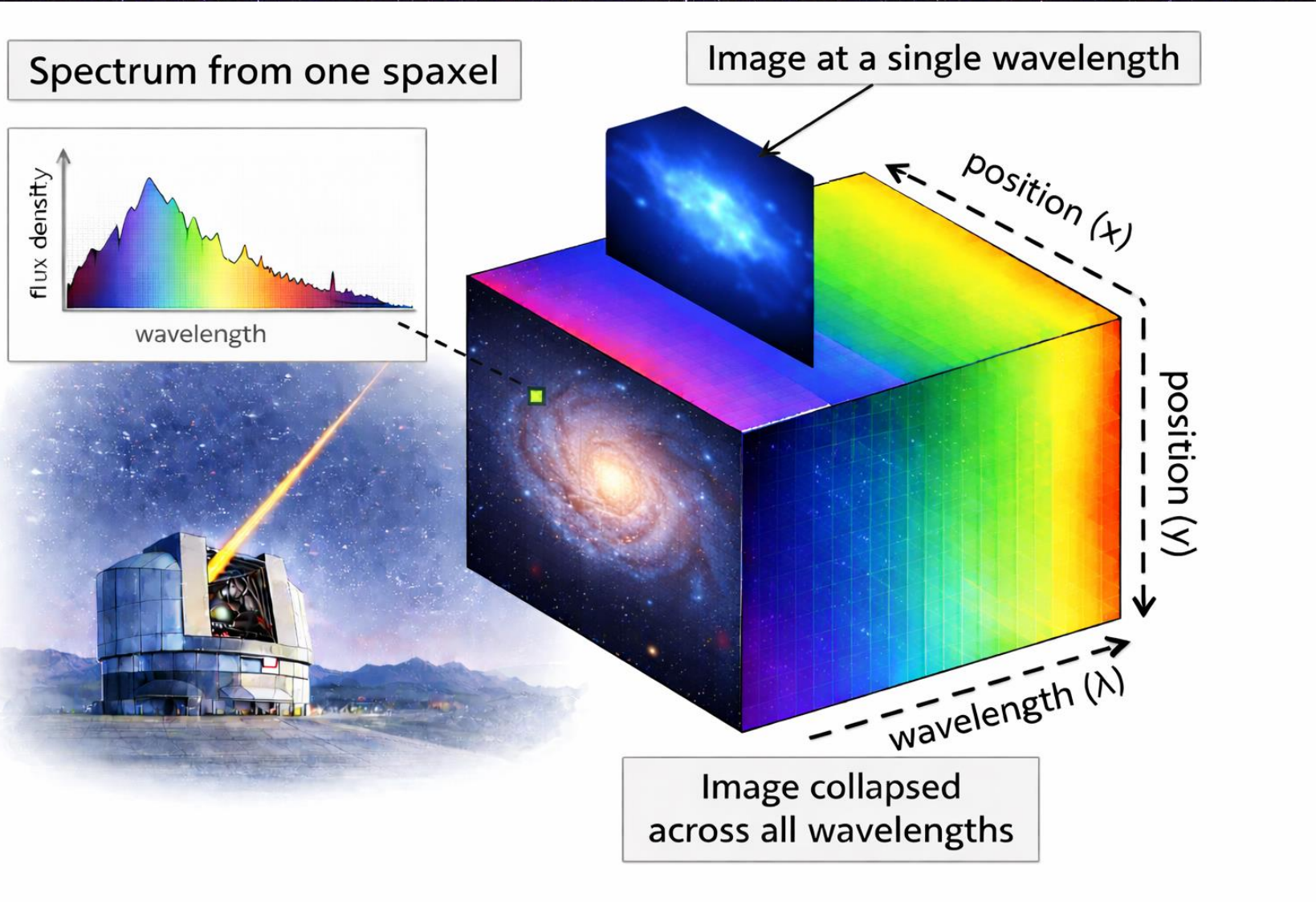


Spatially resolved scales



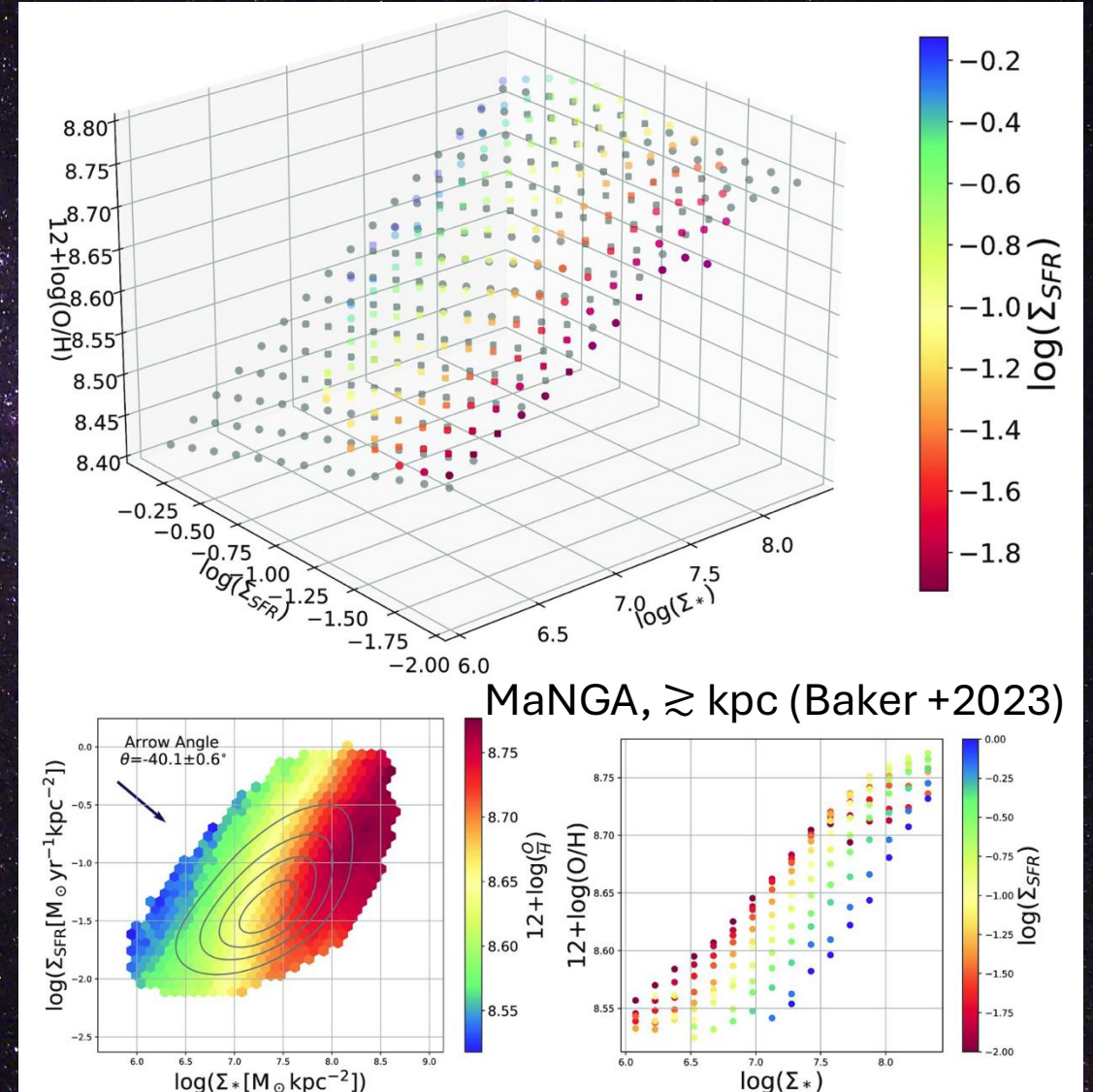
Require Integral Field Unit (IFU) data

Integral Field Unit (IFU) & Datacube



Resolved scaling relations

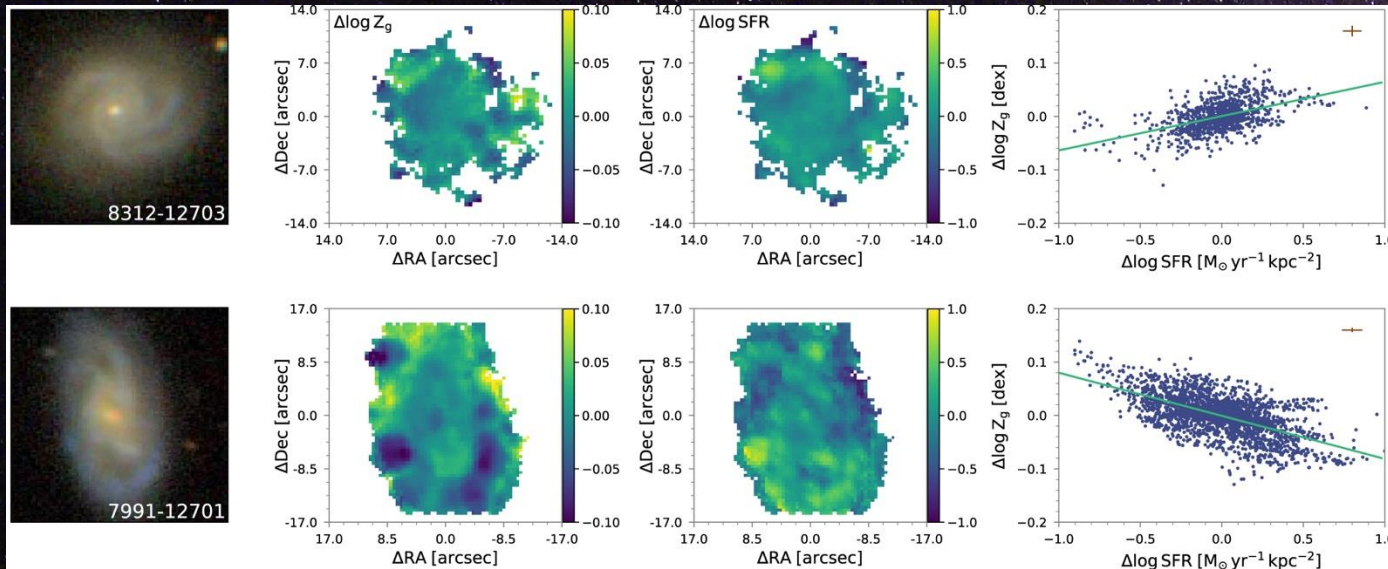
- **rSFMS (bottom left)**: Σ_{SFR} correlates with Σ_*
- **rMZR (bottom right)**: Z_{gas} rises with Σ_* ; steep at low-mass end, saturation at high-mass
- **rFMR (top)**: A 3D surface of $Z_{\text{gas}}(\Sigma_*, \Sigma_{\text{SFR}})$, but **debated!**
 - Some IFU surveys found no clear signal of secondary dependence on Σ_{SFR} (different Z_{gas} diagnostics and sample selection)
 - Still an anti-correlation between $12 + \log(\text{O}/\text{H})$ and Σ_{SFR} at fixed Σ_* ?



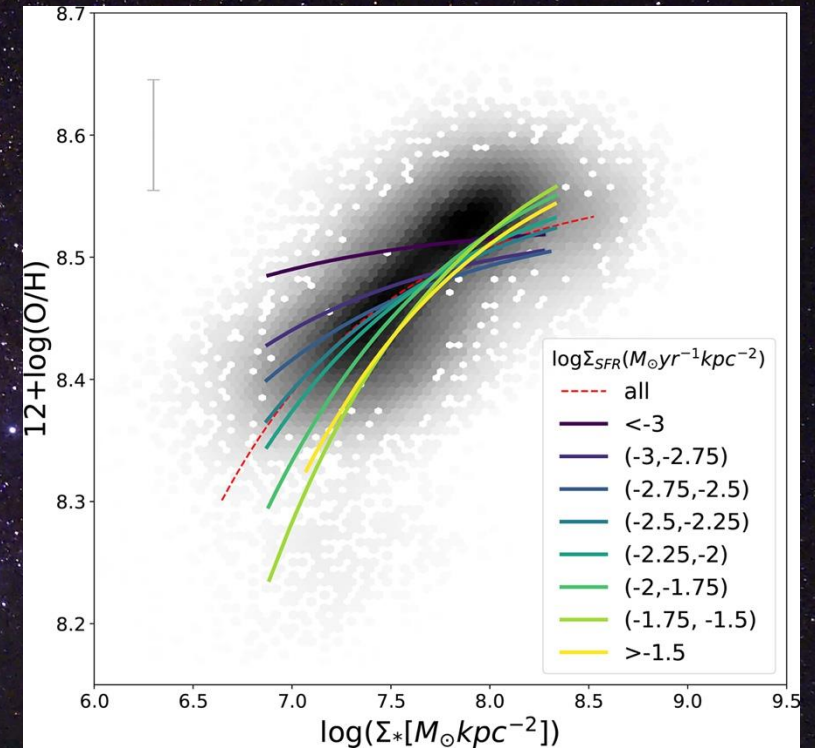
rMZR's secondary dependence on Σ_{SFR}



1. An anti-correlation between Z_{gas} and Σ_{SFR} at fixed Σ_* ?
2. What causes this secondary dependence?



MaNGA, $\gtrsim$ kpc (Sánchez-Menguiano+2019)



MAGPI, $\gtrsim$ kpc (Koller+2024)



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Observational gap

Global scaling relations: well-established in past two decades

Resolved scaling relations (IFU, past decade):

- kpc scales ($\gtrsim$ kpc): MaNGA, SAMI, CALIFA, MUSE-Wide, GASP, MAGPI, etc.
- sub-kpc scales (~ 100 pc)
 - { Field galaxies: PHANGS, MAD.
 - { Cluster galaxies: ??? (this work)



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My PhD goals

Answer the following questions at ~ 100 pc scales:

- Project 1: Resolved scaling relations between Σ_* , Σ_{SFR} , Z_{gas} .
- Project 2: How do environment and other physical conditions affect resolved scaling relations?
- Project 3: Effect of molecular gas on resolved scaling relations.

Understanding this requires resolved spectroscopy across an environmentally processed spiral population



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This Work. MAUVE Survey (MUSE + ALMA)

- 40 late-type Virgo galaxies; VLT/MUSE provides ~ 100 pc datacubes
- Complementary ALMA (molecular gas, down to ~ 60 pc).
- Targets span several infall phases
- Survey current status:
 - MAUVE-MUSE: 26/40 fully delivered (14 used in my first project).
 - MAUVE-ALMA: 34/40 fully delivered



MAUVE

Multiphase Astrophysics to Unveil the Virgo Environment



Pipeline (`Mass.py`): Stellar Mass Map

MUSE Cube

nGIST SFH Maps

BaSTI SSP
(Chabrier IMF)

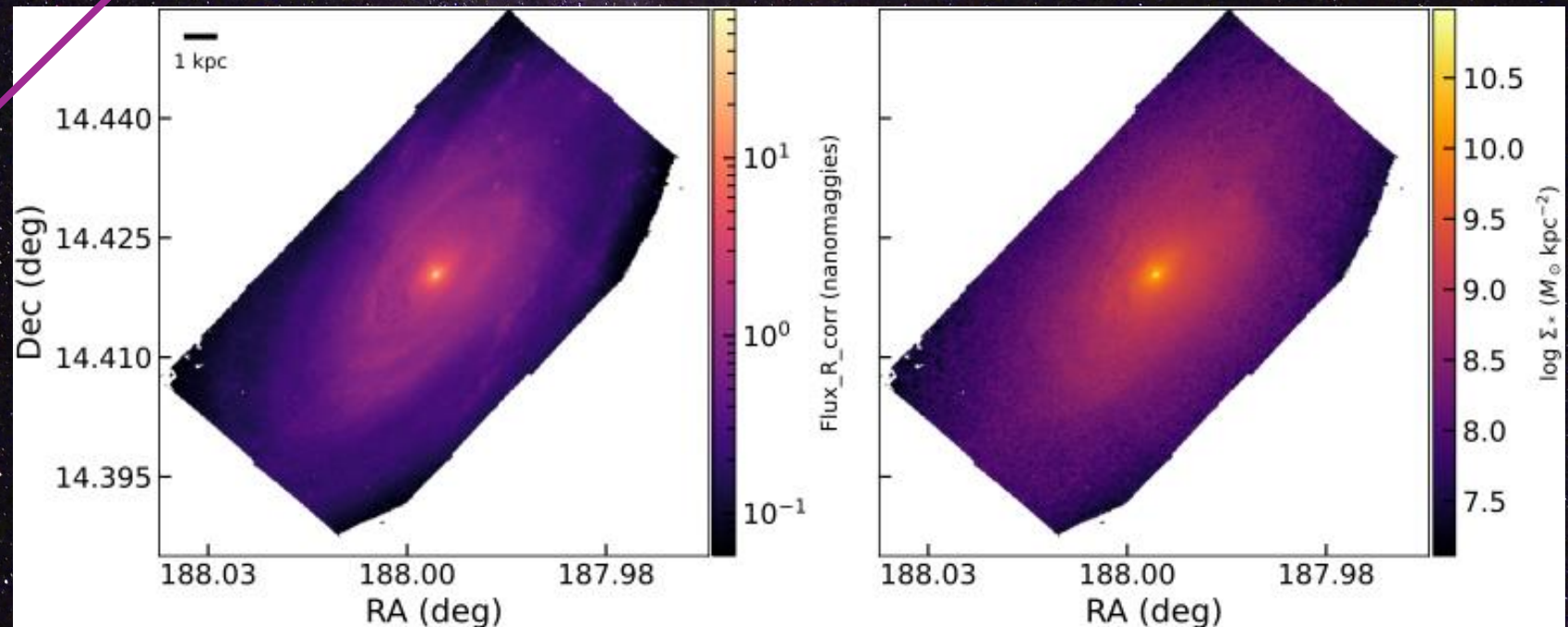
R-band
Photometry

SFH-weighted
M/L Ratio

E(B-V)

Inclination
Correction

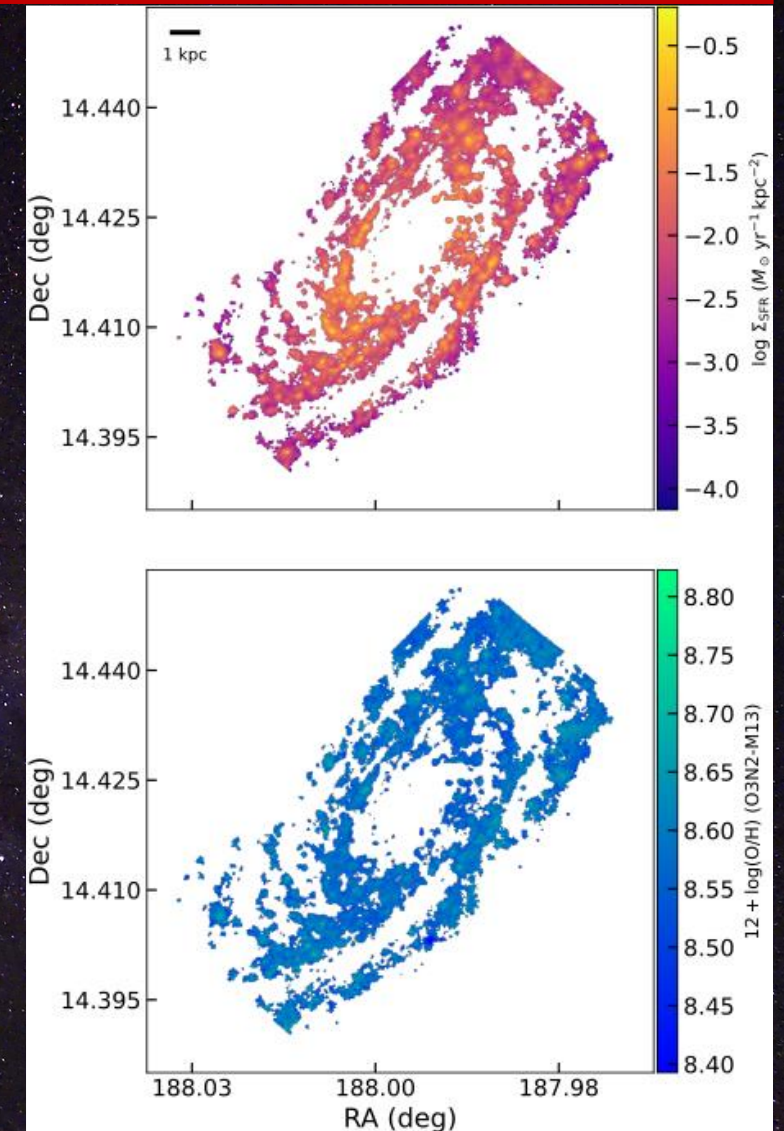
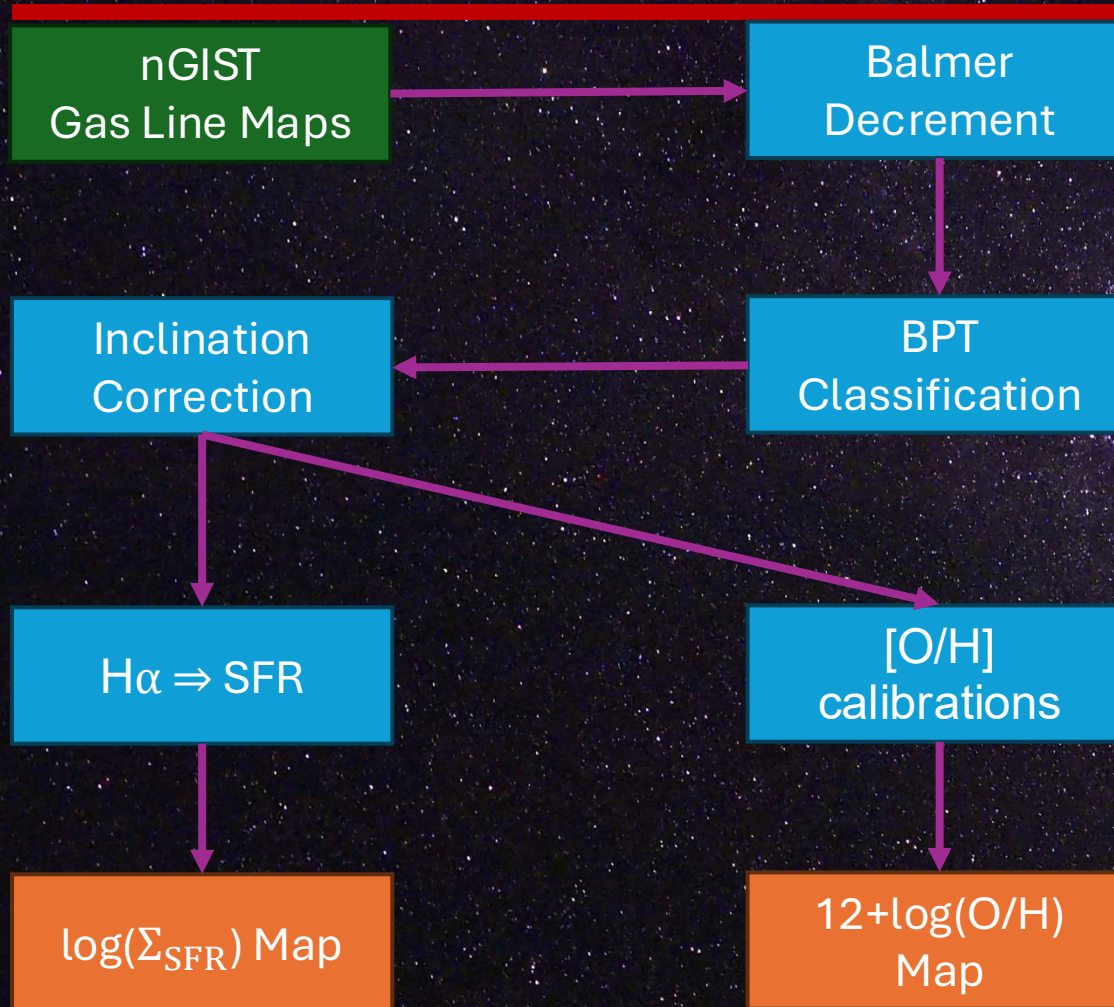
$\log(\Sigma_*)$ Map





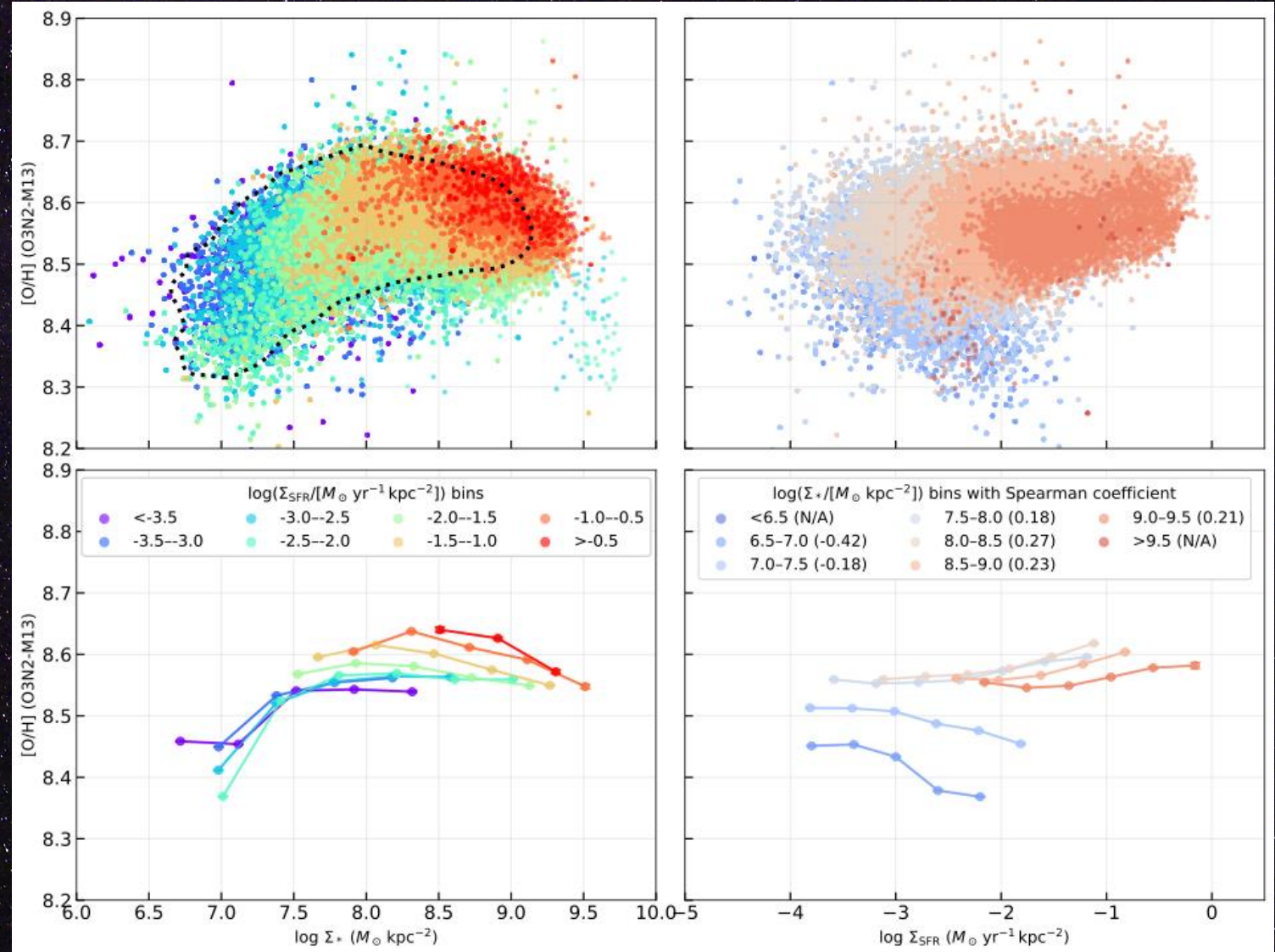
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Pipeline (`SFR+Z.py`): SFR & [O/H] Maps



rMZR and Σ_{SFR} -- Z_{gas} relation

- Left: rMZR, color-coding by Σ_{SFR}
- Right: Σ_{SFR} -- Z_{gas} relation, color-coding by Σ_*
 - Low- Σ_* end: anti-correlated;
 - High- Σ_* end: correlated.





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$\Delta\Sigma_{\text{SFR}}$, ΔZ_{gas} , Local Moran correlation and Σ_* maps

Top panels:

In each Σ_* bin (bottom right), for each spaxel i , we compute

$$\begin{cases} \Delta \log_{10} \Sigma_{\text{SFR}}(i) \equiv \log_{10} \Sigma_{\text{SFR}}(i) - \langle \log_{10} \Sigma_{\text{SFR}} \rangle_m, \\ \Delta [\text{O}/\text{H}](i) \equiv [\text{O}/\text{H}](i) - \langle [\text{O}/\text{H}] \rangle_m, \end{cases}$$

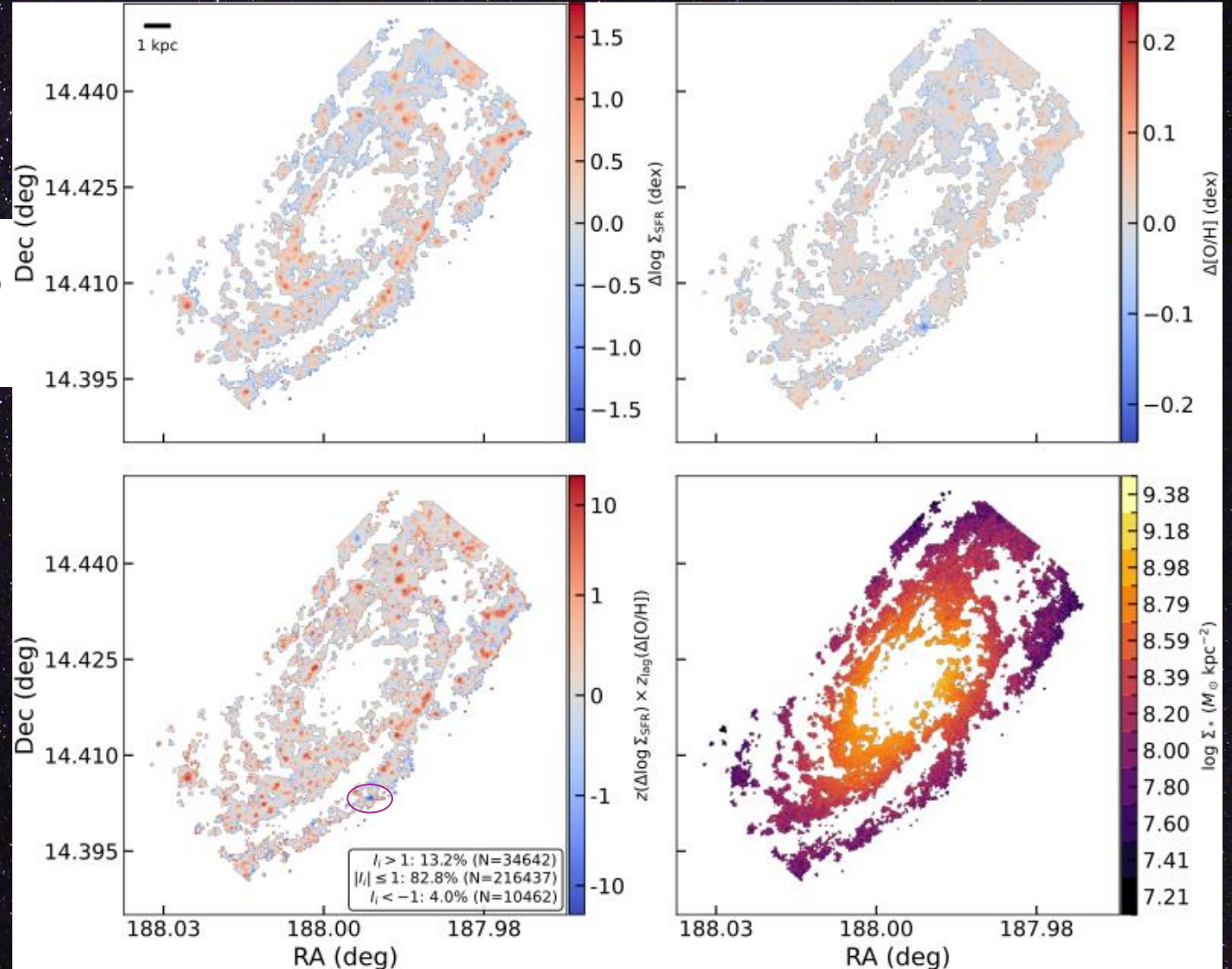
Bottom left:

Local Moran-like correlation Map

Red: correlation

Blue: anti-correlation

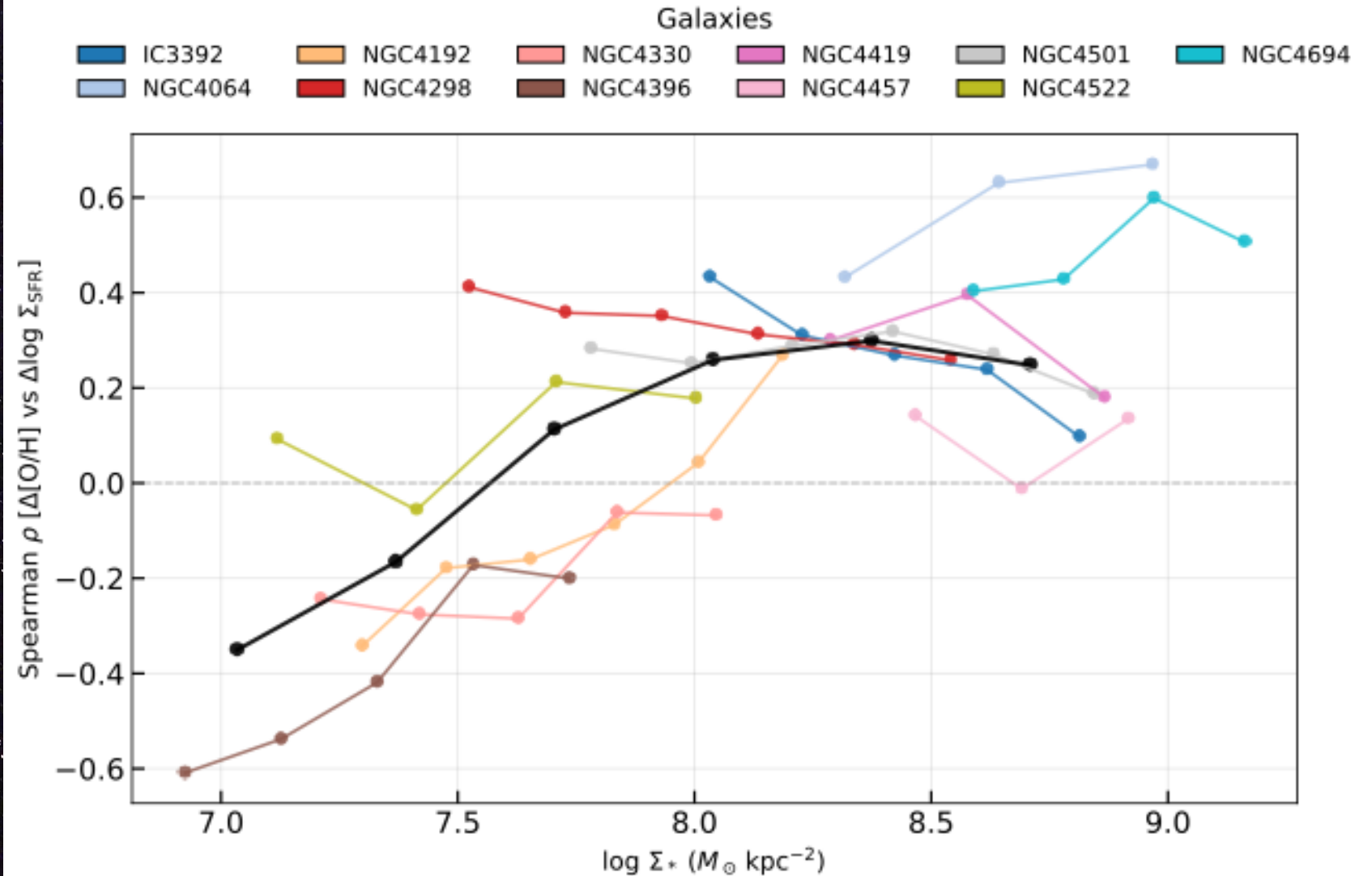
Both correlated and anti-correlated Hii regions are presented everywhere



Mass-dependent reversal

Spearman coefficient ρ
between $\Delta[\text{O}/\text{H}]$ and
 $\Delta \log \Sigma_{\text{SFR}}$, as a function
of $\log \Sigma_*$

Low- Σ_* : anti-correlated
High- Σ_* : correlated
Inversion point:
 $\log(\Sigma_*/M_\odot \text{kpc}^{-2}) \sim 7.5$
– 8.0



What drives this $\Delta\Sigma_{\text{SFR}}--\Delta Z_{\text{gas}}$ (anti-)correlation?

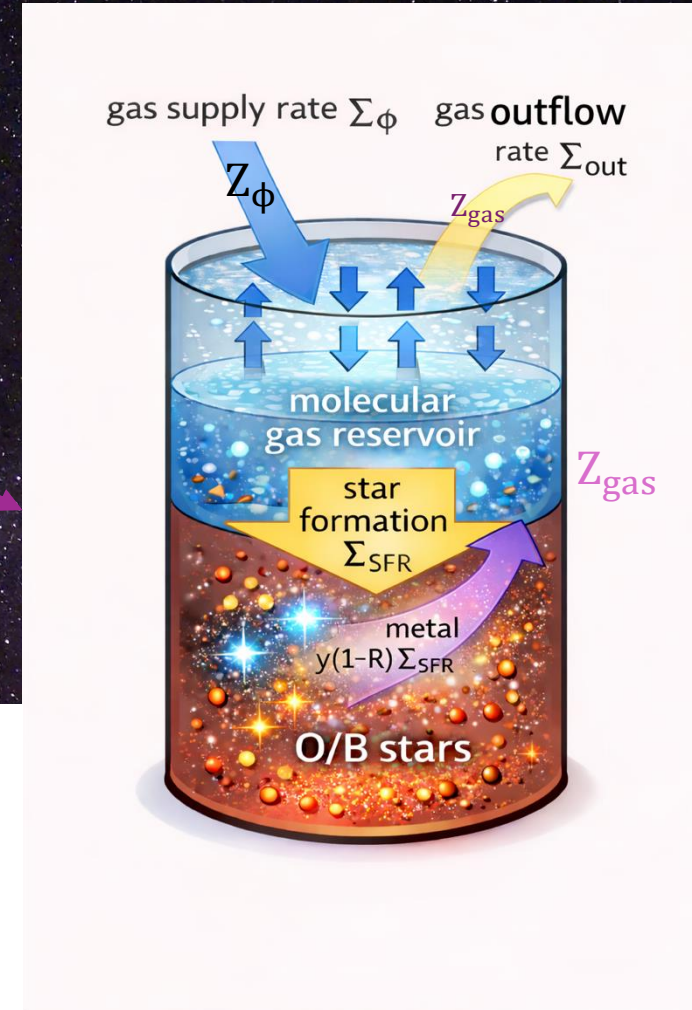


Molecular gas mass and gas-phase metal mass continuity equations:

$$\frac{d\Sigma_g(t)}{dt} = \Sigma_\Phi(t) - (1 - R)\Sigma_{\text{SFR}}(t) - \Sigma_{\text{out}}(t), \quad (1)$$

$$\frac{d\Sigma_Z(t)}{dt} = y(1 - R)\Sigma_{\text{SFR}}(t) - [(1 - R)\Sigma_{\text{SFR}}(t) + \Sigma_{\text{out}}(t)]Z(t) + \Sigma_\Phi(t)Z_\Phi, \quad (2)$$

Spatially resolved gas-regulator model



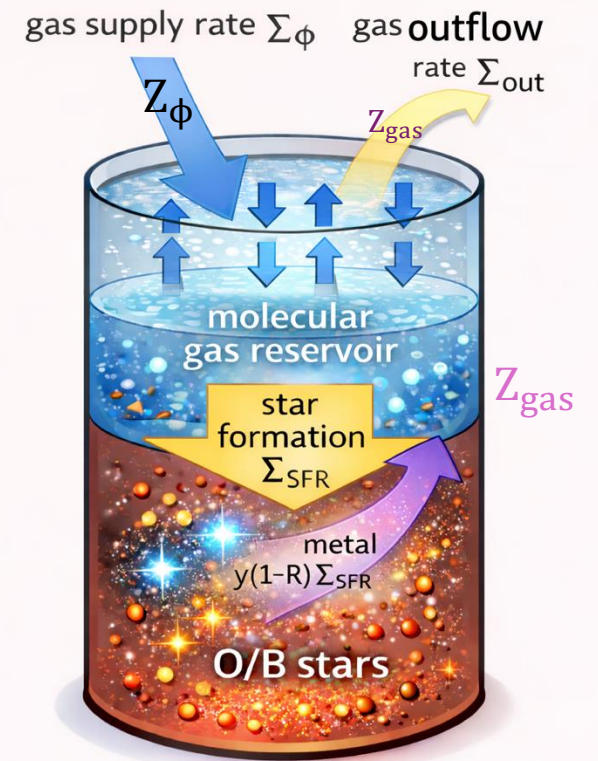
Changes of SFR vs Changes of GSR

$$\delta Z(t) \propto \delta \log_{10}(\Sigma_{\text{SFR}}(t)) - \delta \log_{10}(\Sigma_{\Phi}(t))$$

Star-formation-driven
variability: change of
 Σ_{SFR} in log scales

Gas-supply-driven
variability: change of Σ_{Φ}
in log scales

Star-formation-driven variability dominates $\Rightarrow \uparrow \Sigma_{\text{SFR}}$
(metal yields) with smaller changes in $\Sigma_{\Phi} \Rightarrow \uparrow Z_{\text{gas}}$.
Gas-supply-driven variability dominates $\Rightarrow \uparrow \Sigma_{\text{SFR}}$ with
with larger changes in Σ_{Φ} (Z_{Φ} dilution) $\Rightarrow \downarrow Z_{\text{gas}}$.





Summary of Project 1 (finished)

1. **Derive** my pipelines and analyse 14 MAUVE—MUSE galaxies
2. **Confirm** the rMZR's secondary dependence on Σ_{SFR} at ~ 100 pc scales
3. **Detect** a mass-dependent inversion in residual $\Delta Z_{\text{gas}} - \Delta \Sigma_{\text{SFR}}$ coupling
4. **Show** that this flip-sign is spatially varying in the local Moran-like map
5. **Set up** a spatially resolved gas-regulator model and **interpret** the flip-sign as a transition from gas supply/dilution-dominated to star formation/enrichment-dominated regulation

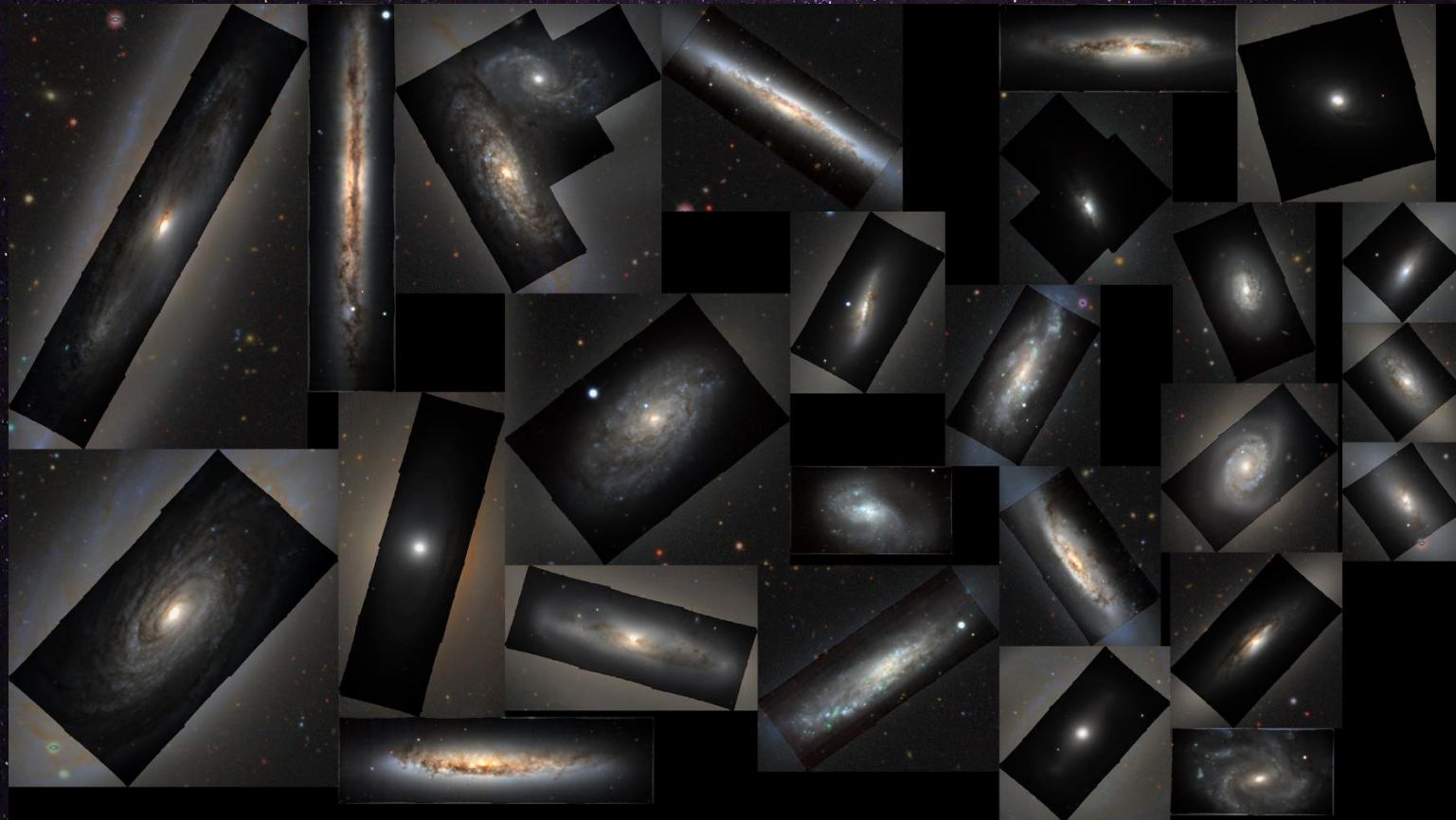
MAUVE—MUSE: When Metallicity Follows or Fights Star Formation—A Mass-Dependent Inversion in Virgo Galaxies

Submitted

Rongjun Huang (黄镭钧)^{1,*}, Luca Cortese¹, Barbara Catinella¹, Luke J. M. Davies¹,
Toby Brown², Andrei Ristea^{3,4}, Alessandro Boselli⁵, Andrew J. Battisti^{1,6}, Vicente Villanueva^{7,8},
Kristine Spekkens⁹, Sara L. Ellison¹⁰, Daniel A. Dale¹¹, Sabine Thater¹², Amirnezhad Amiri^{13,14}

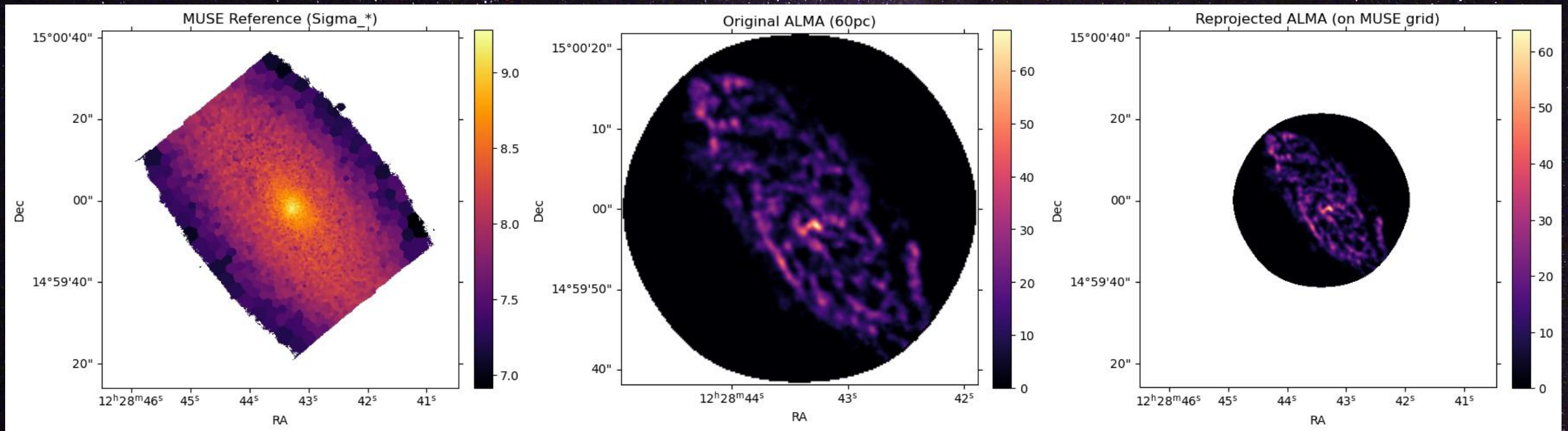
Project 2: 26 MAUVE–MUSE galaxies

1. ~Double our sample sizes: $14 \Rightarrow 26$ MAUVE–MUSE galaxies
2. Robust maps: improved emission-line fitting + more physical parameters
3. Quantify how metallicity is modulated by environment

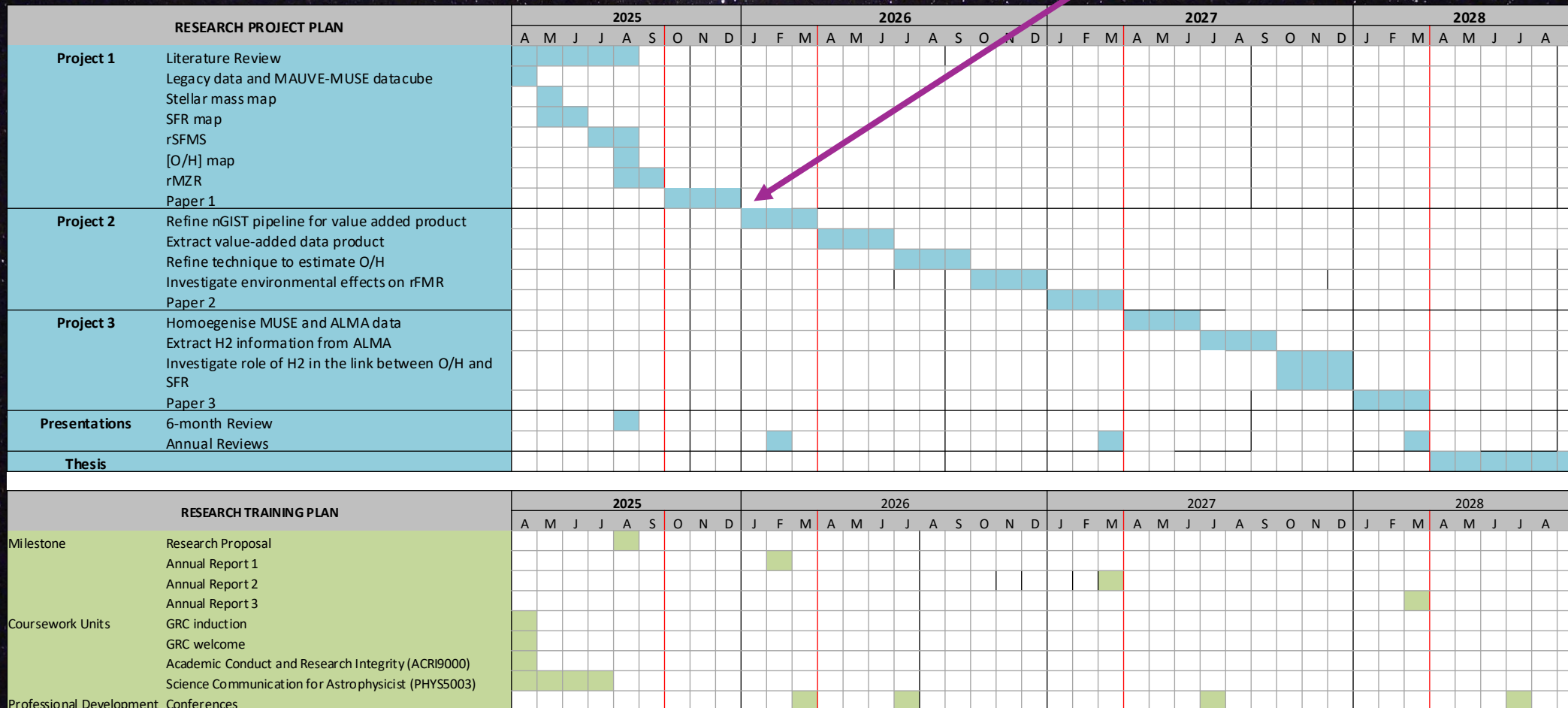


Project 3: Adding MAUVE—ALMA

1. Match MUSE + ALMA: common grid / resolution
2. Measurement of Σ_{gas} and GSR (Σ_{ϕ})
3. Key test: does the $Z_{\text{gas}} - \Sigma_{\text{SFR}}$ behaviour track gas supply



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End

Thank you !
Questions?

